

What next for newly diagnosed glioblastoma?

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Glioblastoma is the most common primary brain tumor in adults. Despite current multimodality treatment including surgical resection and temozolomide-based chemoradiotherapy, median survival is only 14-16 months. Characterization of molecular alterations in glioblastoma has identified prognostic subgroups and therapeutic opportunities for clinical trials across glioblastoma subsets. Following a number of negative Phase III trials testing temozolomide dose intensification and angiogenesis inhibition, recent interim analysis data indicate survival prolongation with use of a device (Optune™) delivering alternating electrical field therapy in newly diagnosed glioblastoma patients. In this review, we present an overview of the data supporting the current standard of care and discuss novel experimental therapies in early and late phase clinical testing including devices, small molecule drugs, angiogenesis inhibitors, oncolytic virotherapy and immunotherapy.

Management of Recurrent Glioblastomas: What Can We Learn from the French Glioblastoma Biobank?

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Safe maximal resection followed by radiotherapy plus concomitant and adjuvant temozolomide (TMZ) is universally accepted as the first-line treatment for glioblastoma (GB), but no standard of care has yet been defined for managing recurrent GB (rGB). We used the French GB biobank (FGB) to evaluate the second-line options currently used, with a view to defining the optimal approach and future directions in GB research. We retrospectively analyzed data for 338 patients with de novo isocitrate dehydrogenase (IDH)-wildtype GB recurring after TMZ chemoradiotherapy. Cox proportional hazards models and Kaplan-Meier analyses were used to investigate survival outcomes. Median overall survival after first surgery (OS1) was 19.8 months (95% CI: 18.5-22.0) and median OS after first progression (OS2) was 9.9 months (95% CI: 8.8-10.8). Two second-line options were noted for rGB patients in the FGB: supportive care and treatments, with systemic treatment being the treatment most frequently used. The supportive care option was independently associated with a shorter OS2 ($p < 0.001$). None of the systemic treatment regimens was unequivocally better than the others for rGB patients. An analysis of survival outcomes based on time to first recurrence (TFR) after chemoradiotherapy indicated that survival was best for patients with a long TFR (≥ 18 months; median OS1: 44.3 months (95% CI: 41.7-56.4) and median OS2: 13.0 months (95% CI: 11.2-17.7), but that such patients constituted only a small proportion of the total patient population (13.0%). This better survival appeared to be more strongly associated with response to first-line treatment than with response to second-line treatment, indicating that the recurring tumors were more aggressive and/or resistant than the initial tumors in these patients. In the face of high rates of treatment failure for GB, the establishment of well-designed large cohorts of primary and rGB samples, with the help of biobanks, such as the FGB, taking into account the TFR and survival outcomes of GB patients, is urgently required for solid comparative biological analyses to drive the discovery of novel prognostic and/or therapeutic clinical markers for GB.

Biophysical and Biological Mechanisms of Tumor Treating Fields in Glioblastoma.

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Glioblastoma (GBM) is one of the most aggressive forms of brain cancer that presents with a median survival rate of 14-30 months and along with a discouraging five-year survival rate of 4-5%. Standard treatment of newly diagnosed GBM, also known as the Stupp protocol, includes a maximally safe surgical resection followed by radiation and chemotherapy. Despite these treatment regimens, recurrence is almost inevitable, emphasizing the need for new therapies to combat the aggressive nature of GBMs. Tumor Treating Fields (TTFs) are a relatively new application to the treatment of GBMs, and results have been promising with both progression-free survival and overall survival when TTFs have been used in combination with temozolomide. This article critically reviews the biophysical and biological mechanisms of TTFs, their clinical efficacy, and discusses the results in clinical trials, including EF-11 and EF-14. Both trials have demonstrated that TTFs can enhance progression free survival and overall survival without compromising quality of life or causing severe adverse effects. Despite the high cost associated with TTFs and the need for further analysis to determine the most effective ways to integrate TTFs into GBM treatments, TTFs represent a significant advancement in GBM therapy and offer hope for improved patient prognosis.

Management of glioblastoma: State of the art and future directions.

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Glioblastoma is the most common malignant primary brain tumor. Overall, the prognosis for patients with this disease is poor, with a median survival of <2 years. There is a slight predominance in males, and incidence increases with age. The standard approach to therapy in the newly diagnosed setting includes surgery followed by concurrent radiotherapy with temozolomide and further adjuvant temozolomide. Tumor-treating fields, delivering low-intensity alternating electric fields, can also be given concurrently with adjuvant temozolomide. At recurrence, there is no standard of care; however, surgery, radiotherapy, and systemic therapy with chemotherapy or bevacizumab are all potential options, depending on the patient's circumstances. Supportive and palliative care remain important considerations throughout the disease course in the multimodality approach to management. The recently revised classification of glioblastoma based on molecular profiling, notably isocitrate dehydrogenase (IDH) mutation status, is a result of enhanced understanding of the underlying pathogenesis of disease. There is a clear need for better therapeutic options, and there have been substantial efforts exploring immunotherapy and precision oncology approaches. In contrast to other solid tumors, however, biological factors, such as the blood-brain barrier and the unique tumor and immune microenvironment, represent significant challenges in the development of novel therapies. Innovative clinical trial designs with biomarker-enrichment strategies are needed to ultimately improve the outcome of patients with glioblastoma.

Temozolomide treatment outcomes and immunotherapy efficacy in brain tumor.

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Glioblastoma (GBM) has a survival rate of around 2 years with aggressive current standard of care. While other tumors have responded favorably to trials combining immunotherapy and chemotherapy, GBM remains uniformly deadly with minimal increases in overall survival. GBM differ from others due to being isolated behind the blood brain barrier, increased heterogeneity and mutational burden, and immunosuppression from the brain environment and tumor itself. We have reviewed clinical and

preclinical studies investigating how different doses (dose intense (DI) and metronomic) and timing of immunotherapy following TMZ treatment can eradicate tumor cells, alter tumor mutational burden, and change immune cells. Recent clinical trials with standard of care (SoC), DI and metronomic TMZ regimes are not able to completely eradicate GBM. Elevated TMZ levels in DI treatment can overcome MGMT resistance but may result in hypermutation of surviving tumor cells. Higher levels of TMZ will also generate a higher degree of lymphopenia compared to SoC and metronomic regimes in preclinical studies. The different levels of lymphopenia and tumor eradication discussed in this review suggest possible beneficial pairings between immunotherapy and TMZ treatment. Treatments resulting in profound lymphopenia will allow for expansion of vaccine specific T cells or of CAR T cells. Clinical and preclinical studies are currently comparing different combinations of TMZ and immunotherapy timing to treat GBM through a balance between tumor killing and immune cell expansion. More frequent immune monitoring time points in ongoing clinical trials are crucial for further development of these combinations.

Treatment of glioblastoma with re-purposed renin-angiotensin system modulators: Results of a phase I clinical trial.

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Glioblastoma is the most common and most aggressive primary brain cancer in adults. Standard treatment of glioblastoma consisting of maximal safe resection, adjuvant radiotherapy and chemotherapy with temozolomide, results in an overall median survival of 14.6 months. The aggressive nature of glioblastoma has been attributed to the presence of glioblastoma stem cells which express components of the renin-angiotensin system (RAS). This phase I clinical trial investigated the tolerability and efficacy of a treatment targeting the RAS and its converging pathways in patients with glioblastoma. Patients who had relapsed following standard treatment of glioblastoma who met the trial criteria were commenced on dose-escalated oral RAS modulators (propranolol, aliskiren, cilazapril, celecoxib, curcumin with piperine, aspirin, and metformin). Of the 17 patients who were enrolled, ten completed full dose-escalation of the treatment. The overall median survival was 19.9 (95% CI:14.1-25.7) months. Serial PET-CTs showed a reduction in both tumor volume and uptake in one patient, an increase in tumor uptake in nine patients with decreased ($n = 1$), unchanged ($n = 1$) and increased ($n = 7$) tumor volume, in the ten patients who had completed full dose-escalation of the treatment. Two patients experienced mild side effects and all patients had preservation of quality of life and performance status during the treatment. There is a trend towards increased survival by 5.3 months although it was not statistically significant. These encouraging results warrant further clinical trials on this potential novel, well-tolerated and cost-effective therapeutic option for patients with glioblastoma.

A phase II open label, randomised study of ipilimumab with temozolomide versus temozolomide alone after surgery and chemoradiotherapy in patients with recently diagnosed glioblastoma: the Ipi-Glio trial protocol.

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Median survival for patients with glioblastoma is less than a year. Standard treatment consists of surgical debulking if feasible followed by temozolomide chemo-radiotherapy. The immune checkpoint inhibitor

ipilimumab targets cytotoxic T-lymphocyte-associated protein 4 (CTLA-4) and has shown clinical efficacy in preclinical models of glioblastoma. The aim of this study is to explore the addition of ipilimumab to standard therapy in patients with glioblastoma. Ipi-Glio is a phase II, open label, randomised study of ipilimumab with temozolomide (Arm A) versus temozolomide alone (Arm B) after surgery and chemoradiotherapy in patients with recently diagnosed glioblastoma. Planned accrual is 120 patients (Arm A: 80, Arm B: 40). Endpoints include overall survival, 18-month survival, 5-year survival, and adverse events. The trial is currently recruiting in seven centres in the United Kingdom. ISRCTN84434175. Registered 12 November 2018.

Therapeutic Immunization against Glioblastoma.

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Glioblastoma is the most common form of brain cancer in adults that produces severe damage to the brain leading to a very poor survival prognosis. The standard of care for glioblastoma is usually surgery, as well as radiotherapy followed by systemic temozolomide chemotherapy, resulting in a median survival time of about 12 to 15 months. Despite these therapeutic efforts, the tumor returns in the vast majority of patients. When relapsing, statistics suggest an imminent death dependent on the size of the tumor, the Karnofsky Performance Status, and the tumor localization. Following the standard of care, the administration of Bevacizumab, inhibiting the growth of the tumor vasculature, is an approved medicinal treatment option approved in the United States, but not in the European Union, as well as the recently approved alternating electric fields (AEFs) generator NovoTTF/Optune. However, it is clear that regardless of the current treatment regimens, glioma patients continue to have dismal prognosis and novel treatments are urgently needed. Here, we describe different approaches of recently developed therapeutic glioma brain cancer vaccines, which stimulate the patient's immune system to recognize tumor-associated antigens (TAA) on cancer cells, aiming to instruct the immune system to eventually attack and destroy the brain tumor cells, with minimal bystander damage to normal brain cells. These distinct immunotherapies may target particular glioma TAAs which are molecularly defined, but they may also target broad patient-derived tumor antigen preparations intentionally evoking a very broad polyclonal antitumor immune stimulation.

[Present status and future prospects of multi-disciplinary therapy for malignant gliomas].

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The prognosis of malignant gliomas is poor. The 5-year survival rate of patients with glioblastomas is still less than 10%. It is difficult to cure this disease by a single treatment modality, and thus, the use of multi-disciplinary therapy mainly comprising surgery, radiotherapy, and chemotherapy is necessary. Malignant gliomas invade the adjacent brain tissue; however, it is well known that greater the extent of tumor resection better is the prognosis. Maximal resection is very important, and various procedures such as intraoperative navigation, evoked potential, photodynamic diagnosis, intraoperative magnetic resonance imaging (MRI), and awake surgery are performed to preserve speech and motor function in patients. According to a phase III trial by the European Organization for Research and Treatment for Cancer (EORTC), the standard therapy for glioblastomas is radiotherapy with concomitant administration

of temozolomide; however, the median survival time is only 14.6 months. Multi-institutional, cooperative trials are necessary for the development of more effective and safe standard therapy.

Personalized Treatment of Glioblastoma: Current State and Future Perspective.

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Glioblastoma (GBM) is the most aggressive glial tumor of the central nervous system. Despite intense scientific efforts, patients diagnosed with GBM and treated with the current standard of care have a median survival of only 15 months. Patients are initially treated by a neurosurgeon with the goal of maximal safe resection of the tumor. Obtaining tissue samples during surgery is indispensable for the diagnosis of GBM. Technological improvements, such as navigation systems and intraoperative monitoring, significantly advanced the possibility of safe gross tumor resection. Usually within six weeks after the surgery, concomitant radiotherapy and chemotherapy with temozolomide are initiated. However, current radiotherapy regimens are based on population-level studies and could also be improved. Implementing artificial intelligence in radiotherapy planning might be used to individualize treatment plans. Furthermore, detailed genetic and molecular markers of the tumor could provide patient-tailored immunochemotherapy. In this article, we review current standard of care and possibilities of personalizing these treatments. Additionally, we discuss novel individualized therapeutic options with encouraging results. Due to inherent heterogeneity of GBM, applying patient-tailored treatment could significantly prolong survival of these patients.
